

# Esmolol: Drug information - UpToDate

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Esmolol: Drug information

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For additional information see "Esmolol: Patient drug information" and "Esmolol: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Brevibloc;
- Brevibloc in NaCl;
- Brevibloc Premixed;
- Brevibloc Premixed DS

Brand Names: Canada

- Brevibloc

Pharmacologic Category

- Antiarrhythmic Agent, Class II;
- Antihypertensive;
- Beta-Blocker, Beta-1 Selective

Dosing: Adult

Acute aortic syndromes/acute aortic dissection

### **Acute aortic syndromes/acute aortic dissection (off-label use):**

**Note:** Manage patients on an emergency basis (including operative assessment) by first controlling pain with IV opioids and heart rate (target 60 to 80 beats per minute) with a parenteral beta blocker (eg, esmolol) while targeting systolic BP of  $\leq 120$  mm Hg (or lowest tolerated pressure without compromising perfusion). If systolic BP remains elevated after heart rate is controlled at 60 to 80 beats per minute with beta-blockade, may consider adding a parenteral vasodilator (ie, nitroprusside or nicardipine). Invasive BP monitoring, preferably via arterial line, in a critical care unit is recommended for appropriate dose titration (Ref).

**IV:** Initial: 500 mcg/kg loading dose over 1 minute, followed by a continuous infusion of 25 to 50 mcg/kg/minute. Titrate infusion by 25 to 50 mcg/kg/minute every 5 minutes as needed to achieve target heart rate and BP; for more rapid BP control, may consider a repeat loading dose (eg, 500 mcg/kg) before every up titration; maximum continuous infusion dose: 300 mcg/kg/minute (Ref).

Atrial fibrillation/flutter, acute ventricular rate control

**Atrial fibrillation/flutter, acute ventricular rate control:**

**Note:** Avoid in patients with decompensated heart failure; for unstable patients, electrical cardioversion is preferred (Ref).

Loading doses (optional): **IV:** 500 mcg/kg over 1 minute followed by a continuous infusion; may administer repeat bolus doses of 500 mcg/kg prior to each increase in continuous infusion rate in order to achieve a more rapid response (Ref).

Continuous infusion: **IV:** Initial: 50 mcg/kg/minute; for inadequate response, may increase in increments of 50 mcg/kg/minute at  $\geq 4$ -minute intervals up to a maximum of 300 mcg/kg/minute. To achieve a more rapid response, administer a repeat bolus before increasing the continuous infusion rate. In the absence of a bolus, the effects of continuous infusion rate changes may not be evident for up to 30 minutes (Ref).

Hypertensive emergency

**Hypertensive emergency (off-label use):**

**Note :** When used for hypertensive emergency without a compelling condition, in general, reduce mean arterial BP gradually by  $\sim 10\%$  to  $20\%$  over the first hour. If stable, may then reduce to a goal BP of  $<160/100$  mm Hg for the next 2 to 6 hours, then to 130 to 140 mm Hg over the next 24 to 48 hours to limit target organ injury. Invasive BP monitoring is recommended for appropriate dose titration (Ref).

**IV:** Initial: 250 to 500 mcg/kg loading dose over 1 minute, followed by a continuous infusion of 25 to 50 mcg/kg/minute. Titrate infusion by 25 to 50 mcg/kg/minute every 5 minutes as needed to achieve target BP while maintaining adequate heart rate; for more rapid BP control, may consider a repeat loading dose (eg, 250 to 500 mcg/kg) before every up-titration; maximum continuous infusion dose: 300 mcg/kg/minute (Ref).

Intraoperative and postoperative tachycardia and/or hypertension

**Intraoperative and postoperative tachycardia and/or hypertension:**

*Immediate control:* **IV:** Initial bolus: 1,000 mcg/kg over 30 seconds, followed by a 150 mcg/kg/minute continuous infusion, if necessary. Adjust infusion rate up to a maximum dose of 300 mcg/kg/minute as needed to maintain desired heart rate and/or BP.

*Gradual control:* **IV:** Initial bolus: 500 mcg/kg over 1 minute, followed by a 50 mcg/kg/minute continuous infusion for at least 4 minutes. If clinical response is inadequate, continuous infusion may be titrated upward in 50 mcg/kg/minute increments (increased no more frequently than every 4 minutes) to a maximum dose of 300 mcg/kg/minute; may administer an optional 500 mcg/kg loading dose over 1 minute prior to each increase in infusion rate.

**Note:** For control of tachycardia, maintenance doses  $>200$  mcg/kg/minute provide minimal additional effect and are not recommended due to increased adverse effects. For control of postoperative hypertension, higher maintenance doses (eg, 250 to 300 mcg/kg/minute) may be required to control BP.

Sinus tachycardia, inappropriate, noncompensatory

**Sinus tachycardia, inappropriate, noncompensatory:**

**Note:** Treatment should be directed at the underlying cause; if treatment of the underlying cause is ineffective, may consider use of a beta blocker in symptomatic patients (Ref).

Loading doses (optional): **IV:** 500 mcg/kg over 1 minute followed by a continuous infusion; may administer repeat bolus doses of 500 mcg/kg prior to each increase in continuous infusion rate in order to achieve a more rapid response (Ref).

Continuous infusion: **IV:** Initial: 50 mcg/kg/minute; for inadequate response, may increase in increments of 50 mcg/kg/minute at  $\geq 4$ -minute intervals up to a maximum of 300 mcg/kg/minute. To achieve a more rapid response, administer a repeat bolus before increasing the continuous infusion

rate. In the absence of a bolus, the effects of continuous infusion rate changes may not be evident for up to 30 minutes (Ref).

Supraventricular tachycardia

**Supraventricular tachycardia (ie, atrioventricular nodal reentrant tachycardia, atrioventricular reentrant tachycardia, ectopic atrial tachycardia, multifocal atrial tachycardia):**

**Note:** Consider for hemodynamically stable patients if vagal maneuvers and/or adenosine are unsuccessful. Do **not** use in patients with pre-excited atrial fibrillation or suspected wide complex tachycardia associated with an accessory pathway as this can lead to ventricular arrhythmias (Ref).

Loading doses (optional): **IV:** 500 mcg/kg over 1 minute, followed by a continuous infusion; may administer repeat bolus doses of 500 mcg/kg prior to each increase in continuous infusion rate in order to achieve a more rapid response (Ref).

Continuous infusion: **IV:** Initial: 50 mcg/kg/minute; for inadequate response, may increase in increments of 50 mcg/kg/minute at  $\geq 4$ -minute intervals up to a maximum of 300 mcg/kg/minute. To achieve a more rapid response, administer a repeat bolus before increasing the continuous infusion rate. In the absence of a bolus, the effects of continuous infusion rate changes may not be evident for up to 30 minutes (Ref).

Thyroid storm

**Thyroid storm (alternative agent) (off-label use):**

**Note:** For control of adrenergic symptoms until thyroid storm has resolved (Ref). Administer while under continuous cardiac monitoring; may be used if it is uncertain that a beta-blocker will be tolerated since after discontinuation, duration of action is very short. Avoid use in patients with decompensated low-output heart failure (systolic dysfunction) (Ref).

**IV:** 250 to 500 mcg/kg loading dose, followed by a continuous infusion of 50 to 100 mcg/kg/minute; adjust dose based on heart rate and BP. Use in combination with other appropriate agents (Ref).

Ventricular tachycardia

**Ventricular tachycardia (off-label use):** **IV:** 500 mcg/kg bolus followed by 50 mcg/kg/minute (Ref).

**Transition to alternative IV or oral cardiovascular therapies:** Manufacturer general recommendations: Carefully consider the labeling instructions of the selected alternative IV or oral agent, then reduce continuous infusion by 50% thirty minutes following the first dose of the alternative agent. Following the second dose of the alternative agent, patient's response should be monitored and if control is adequate for the first hour, esmolol may be discontinued.

**Dosage adjustment for concomitant therapy:** Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

**IV:** No dosage adjustment necessary. Not removed by hemo- or peritoneal dialysis. Supplemental dose is not necessary.

Dosing: Liver Impairment: Adult

No dosage adjustment necessary.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Esmolol: Pediatric drug information")

**Note:** Dose must be titrated to individual response and tolerance.

Hypertension, acute severe with significant, life-threatening symptoms

**Hypertension, acute severe with significant, life-threatening symptoms (eg, seizures):**  
Limited data available:

Infants, Children, and Adolescents: Continuous IV infusion: 100 to 500 mcg/kg/minute infusion (Ref); another approach is to initiate therapy with a bolus of 100 to 500 mcg/kg over 1 minute, followed by an infusion of 25 to 100 mcg/kg/minute; titrate as needed up to 500 mcg/kg/minute (Ref).

Hypertension, postoperative

**Hypertension, postoperative (congenital heart disease):** Limited data available; large effective dose range reported:

Infants and Children: Initial IV bolus: 100 to 500 mcg/kg over 1 minute, followed by continuous IV infusion: Initial rate: 100 to 500 mcg/kg/minute; titrate to effect; range of effective doses: 125 to 1,000 mcg/kg/minute; higher doses may be needed in patients after repair of coarctation of the aorta (Ref).

Supraventricular tachycardia

**Supraventricular tachycardia (SVT):** Limited data available:

Children and Adolescents: Initial IV bolus: 100 to 500 mcg/kg over 1 minute followed by a continuous IV infusion: Initial rate: 25 to 100 mcg/kg/minute; titrate by 25 to 50 mcg/kg/minute; usual maintenance dose: 50 to 500 mcg/kg/minute (Ref); doses up to 1,000 mcg/kg/minute have been reported (Ref).

**Dosage adjustment for concomitant therapy:** Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Infants, Children, and Adolescents: No dosage adjustment necessary. Not removed by hemodialysis, peritoneal dialysis, or CRRT; supplemental dose is not necessary (Ref).

Dosing: Liver Impairment: Pediatric

There are no pediatric-specific recommendations; based on experience in adult patients, no dosage adjustment required.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%: Cardiovascular: Hypotension (asymptomatic: 25%; symptomatic: 12%)

1% to 10%:

Cardiovascular: Peripheral ischemia (1%)

Gastrointestinal: Nausea (7%), vomiting (1%)

Local: Infusion-site reaction (8%; including induration, irritation, local inflammation, local thrombophlebitis, local tissue necrosis, skin blister)

Nervous system: Agitation (2%), confusion (2%), dizziness ( $\leq 3\%$ ), drowsiness (3%), headache (2%)

<1%:

Cardiovascular: Flushing, syncope

Dermatologic: Pallor

Gastrointestinal: Abdominal distress, constipation, dyspepsia, xerostomia

Genitourinary: Urinary retention

Nervous system: Anxiety, changes in thinking, depression, paresthesia, seizure, speech disturbance

Frequency not defined:

Cardiovascular: Bradycardia (including severe bradycardia), heart block, increased heart rate (moderate increase above pretreatment levels 30 minutes after discontinuation), sinus pause

Endocrine & metabolic: Hyperkalemia

Renal: Renal tubular acidosis (hyperkalemic)

Postmarketing:

Cardiovascular: Coronary artery vasospasm

Dermatologic: Psoriasis, urticaria

Hypersensitivity: Angioedema

Contraindications

Hypersensitivity to esmolol or any component of the formulation; severe sinus bradycardia; heart block greater than first degree or sick sinus syndrome (except in patients with a functioning artificial ventricular pacemaker); cardiogenic shock; decompensated heart failure; IV administration of calcium channel blockers (eg, verapamil) in close proximity to esmolol (ie, while cardiac effects of other drug are still present); pulmonary hypertension

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

*Canadian labeling:* Additional contraindications (not in US labeling): Patients requiring inotropic agents and/or vasopressors to maintain cardiac output and systolic blood pressure; hypotension; right ventricular failure secondary to pulmonary hypertension; untreated pheochromocytoma

Warnings/Precautions

**Concerns related to adverse effects:**

- Anaphylactic reactions: Use caution with history of severe anaphylaxis to allergens; patients taking beta-blockers may become more sensitive to repeated challenges. Treatment of anaphylaxis (eg, epinephrine) in patients taking beta-blockers may be ineffective or promote undesirable effects.
- Extravasation: Vesicant; ensure proper needle or catheter placement prior to and during infusion. Avoid extravasation. Extravasation can lead to skin necrosis and sloughing; avoid infusions into small veins or through a butterfly catheter.
- Hyperkalemia: Esmolol has been associated with elevations in serum potassium and development of hyperkalemia especially in patients with risk factors (eg, kidney impairment); monitor serum potassium during therapy.
- Hypotension: Can commonly occur; patients need close blood pressure monitoring. If an unacceptable drop in blood pressure occurs, reduction in dose or discontinuation may reverse hypotension (usually within 30 minutes).

**Disease-related concerns:**

- Bronchospastic disease: In general, patients with bronchospastic disease should not receive beta-blockers; however, esmolol, with B<sub>1</sub> selectivity, has been used cautiously with close monitoring.
- Cardiovascular disease: Use with caution in patients with compensated heart failure and monitor for a worsening of the condition. Use is contraindicated in patients with decompensated heart failure.
- Conduction abnormality: Can cause bradycardia including sinus pause, heart block, severe bradycardia, and cardiac arrest. Consider preexisting conditions such as first degree AV block, sick sinus syndrome, or other conduction disorders before initiating; use is contraindicated in patients with sick sinus syndrome or second- or third-degree AV block (except in patients with a functioning artificial ventricular pacemaker).
- Diabetes: Use with caution in patients with diabetes mellitus; may potentiate hypoglycemia and/or mask signs and symptoms.

- Kidney impairment: Use with caution in patients with kidney impairment; active metabolite retained.
- Myasthenia gravis: Use with caution in patients with myasthenia gravis.
- Peripheral vascular disease (PVD) and Raynaud disease: Can precipitate or aggravate symptoms of arterial insufficiency in patients with PVD and Raynaud disease. Use with caution and monitor for progression of arterial obstruction.
- Pheochromocytoma (untreated): Adequate alpha-blockade is required prior to use of any beta-blocker.
- Thyroid disease: May mask signs of hyperthyroidism (eg, tachycardia). If hyperthyroidism is suspected, carefully manage and monitor; abrupt withdrawal may exacerbate symptoms of hyperthyroidism or precipitate thyroid storm.
- Vasospastic angina: Beta-blockers without alpha1-adrenergic receptor blocking activity should be avoided in patients with vasospastic angina since unopposed alpha1-adrenergic receptors mediate coronary vasoconstriction and can worsen anginal symptoms (Mayer 1998).

### ***Special populations:***

- Older patients: Bradycardia may be observed more frequently in patients >65 years of age; dosage reductions may be necessary.

### ***Dosage form specific issues:***

- Propylene glycol: Some dosage forms may contain propylene glycol; large amounts are potentially toxic and have been associated with hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Zar 2007). See manufacturer's labeling.

### ***Other warnings/precautions:***

- Abrupt withdrawal: Beta-blocker therapy should not be withdrawn abruptly (particularly in patients with CAD), but gradually tapered to avoid acute tachycardia, hypertension, and/or ischemia. Severe exacerbation of angina, ventricular arrhythmias, and myocardial infarction (MI) have been reported following abrupt withdrawal of beta-blocker therapy. Temporary but prompt resumption of beta-blocker therapy may be indicated with worsening of angina or acute coronary insufficiency.
- Hypertension associated with hypothermia: Use esmolol with caution in patients with hypertension associated with hypothermia; monitor vital signs closely and titrate esmolol slowly.
- Hypovolemic patients: Avoid use in patients with hypovolemia; treat hypovolemia first, otherwise, use of esmolol may attenuate reflex tachycardia and further increase the risk of hypotension.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous, as hydrochloride:

Brevibloc: 100 mg/10 mL (10 mL)

Brevibloc in NaCl: 2000 mg (100 mL); 2500 mg (250 mL)

Generic: 2000 mg (100 mL); 2500 mg (250 mL)

Solution, Intravenous, as hydrochloride [preservative free]:

Brevibloc in NaCl: 2000 mg (100 mL); 2500 mg (250 mL) [latex free, pvc free]

Brevibloc Premixed: 2500 mg (250 mL) [latex free, pvc free]

Brevibloc Premixed DS: 2000 mg (100 mL) [latex free, pvc free]

Generic: 2000 mg (100 mL); 2500 mg (250 mL); 100 mg/10 mL (10 mL); 2500 mg/250 mL (250 mL); 2000 mg/100 mL (100 mL)

Generic Equivalent Available: US

Yes

Pricing: US

**Solution** (Brevibloc in NaCl Intravenous)

2000 mg/100 mL (per mL): \$2.16

2500 mg/250 mL (per mL): \$0.62

**Solution** (Brevibloc Intravenous)

100 mg/10 mL (per mL): \$2.56

**Solution** (Brevibloc Premixed DS Intravenous)

2000 mg/100 mL (per mL): \$1.92

**Solution** (Brevibloc Premixed Intravenous)

2500 mg/250 mL (per mL): \$0.62

**Solution** (Esmolol HCl Intravenous)

100 mg/10 mL (per mL): \$0.88 - \$2.16

2000 mg/100 mL (per mL): \$1.55

2500 mg/250 mL (per mL): \$0.57

**Solution** (Esmolol HCl-Sodium Chloride Intravenous)

2000 mg/100 mL (per mL): \$0.77 - \$5.44

2500 mg/250 mL (per mL): \$0.35 - \$2.02

**Disclaimer:** A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous, as hydrochloride:

Brevibloc: 2500 mg (250 mL); 100 mg/10 mL (10 mL)

Administration: Adult

**IV:** Loading and bolus doses may be administered over 30 to 60 seconds depending on how urgent the need for effect. Infusion into small veins or through a butterfly catheter should be avoided (can cause thrombophlebitis). Medication port of premixed bags should be used to withdraw only the initial bolus, if necessary (not to be used for withdrawal of additional bolus doses).

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation.

**Extravasation management:** If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, remove needle/cannula; elevate extremity; apply dry, warm compresses; may consider hyaluronidase for refractory peripheral extravasations in addition to supportive management (Ref).

*Hyaluronidase (if appropriate): Intradermal or SUBQ:* Inject a total of 1 mL (15 units/mL) as five separate 0.2 mL injections (using a tuberculin syringe) around the site of extravasation; if IV catheter remains in place, administer initial dose intravenously through the infiltrated catheter; may repeat in 30 to 60 minutes if there is no resolution (Ref).

Usual Infusion Concentrations: Adult

**Note:** Premixed solutions available.

**IV infusion:** 2500 mg in 250 mL (concentration: 10,000 **mcg**/mL) **or** 2000 mg in 100 mL (concentration: 20,000 **mcg**/mL) of D<sub>5</sub>W or NS

Administration: Pediatric

**Parenteral:**

**IV:** Commercially available concentrations (10 mg/mL and 20 mg/mL) are iso-osmotic and can be used for direct IV use; administer loading doses over 1 minute. Do not introduce additives into the premixed bags. Medication port of premixed bag should be used to withdraw only the initial bolus; do not use medication port to withdraw additional bolus doses (sterility cannot be assured). ASHP recommends standard concentrations of 10 mg/mL and 20 mg/mL in pediatric patients (Ref).

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation. If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do NOT flush the line; gently aspirate extravasated solution; remove needle/cannula unless needed for antidote; elevate extremity; apply dry, warm compresses; may consider hyaluronidase for refractory peripheral extravasations in addition to supportive management (Ref).

Usual Infusion Concentrations: Pediatric

**Note:** Premixed solutions available.

**IV infusion:** 10 mg/mL, 20 mg/mL

Vesicant/Extravasation Risk

Vesicant

Storage/Stability

Store at 25°C (77°F); excursions permitted to 15°C to 30°C (59°F to 86°F); do not freeze. Protect from excessive heat. Stable for at least 24 hours (under refrigeration or at controlled room temperature) at a final concentration of 10 mg/mL in D5W, D5LR, D5R, D5<sup>1</sup>/<sub>2</sub>NS, D5NS, LR, <sup>1</sup>/<sub>2</sub>NS, or NS. Do not remove overwrap from premixed bags until ready to use; use within 24 hours once medication is initially withdrawn from ready to use bag.

Use: Labeled Indications

**Atrial fibrillation/flutter, acute ventricular rate control:** Control of ventricular rate in patients with atrial fibrillation/flutter

**Intraoperative and postoperative tachycardia and/or hypertension:** Treatment of intraoperative and postoperative tachycardia and/or hypertension

**Sinus tachycardia, inappropriate, noncompensatory:** Treatment of noncompensatory sinus tachycardia

**Supraventricular tachycardia:** Control of ventricular rate in patients with supraventricular tachycardia

Use: Off-Label: Adult



Acute aortic syndromes/acute aortic dissection; Hypertensive emergency; Thyroid storm; Ventricular tachycardia

#### Medication Safety Issues

Sound-alike/look-alike issues:

Esmolol may be confused with Osmitrol.

Brevibloc may be confused with Brevital, Bumex, Buprenex.

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (adrenergic antagonist, IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

#### Metabolism/Transport Effects

None known.

#### Drug Interactions

(For additional information: Launch drug interactions program)

**Note:** Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acetylcholinesterase Inhibitors: May increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Alpha2-Agonists: Beta-Blockers may increase rebound hypertensive effects of Alpha2-Agonists. This effect can occur when the Alpha2-Agonist is abruptly withdrawn. Alpha2-Agonists may increase AV-blocking effects of Beta-Blockers. Sinus node dysfunction may also be enhanced. Management: Closely monitor heart rate during treatment with a beta blocker and clonidine. Withdraw beta blockers several days before clonidine withdrawal when possible, and monitor blood pressure closely. Recommendations for other alpha2-agonists are unavailable. *Risk D: Consider Therapy Modification*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amiodarone: May increase bradycardic effects of Beta-Blockers. Possibly to the point of cardiac arrest. Amiodarone may increase serum concentrations of Beta-Blockers. *Risk C: Monitor*

Antidiabetic Agents: Beta-Blockers (Beta1 Selective) may increase adverse/toxic effects of Antidiabetic Agents. Specifically, beta-blockers may mask the hypoglycemic symptoms of antidiabetic agents. *Risk C: Monitor*

Antipsychotic Agents (Phenothiazines): May increase hypotensive effects of Beta-Blockers. Beta-Blockers may decrease metabolism of Antipsychotic Agents (Phenothiazines). Antipsychotic Agents (Phenothiazines) may decrease metabolism of Beta-Blockers. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Beta2-Agonists: Beta-Blockers (Beta1 Selective) may decrease bronchodilatory effects of Beta2-Agonists. *Risk C: Monitor*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Brigatinib: May decrease antihypertensive effects of Antihypertensive Agents. Brigatinib may increase bradycardic effects of Antihypertensive Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Cafedrine: May increase bradycardic effects of Beta-Blockers. Beta-Blockers may decrease therapeutic effects of Cafedrine. *Risk C: Monitor*

Calcium Channel Blockers (Nondihydropyridine): May increase bradycardic effects of Esmolol. Management: Administration of IV verapamil or diltiazem together with esmolol is contraindicated if one agent is given while the effects of the other are still present. Canadian esmolol labeling specifies that use within 24 hours is contraindicated. *Risk D: Consider Therapy Modification*

Cannabis: Beta-Blockers may increase adverse/toxic effects of Cannabis. Specifically, the risk of hypoglycemia may be increased. *Risk C: Monitor*

Ceritinib: Bradycardia-Causing Agents may increase bradycardic effects of Ceritinib. Management: If this combination cannot be avoided, monitor patients for evidence of symptomatic bradycardia, and closely monitor blood pressure and heart rate during therapy. *Risk D: Consider Therapy Modification*

Cholinergic Agonists: Beta-Blockers may increase adverse/toxic effects of Cholinergic Agonists. Of particular concern are the potential for cardiac conduction abnormalities and bronchoconstriction. *Risk C: Monitor*

Dexmethylphenidate: May decrease therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Dipyridamole: May increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

Disopyramide: May increase bradycardic effects of Beta-Blockers. Beta-Blockers may increase negative inotropic effects of Disopyramide. *Risk C: Monitor*

DOBUtamine: Beta-Blockers may decrease therapeutic effects of DOBUtamine. *Risk C: Monitor*

Dronedarone: May increase bradycardic effects of Beta-Blockers. Dronedarone may increase serum concentrations of Beta-Blockers. This likely applies only to those agents that are metabolized by CYP2D6. Management: Use lower initial beta-blocker doses; adequate tolerance of the combination, based on ECG findings, should be confirmed prior to any increase in beta-blocker dose. Increase monitoring for clinical response and adverse effects. *Risk D: Consider Therapy Modification*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

EPHEDrine (Systemic): Beta-Blockers may decrease therapeutic effects of EPHEDrine (Systemic). *Risk C: Monitor*

EPINEPHrine (Systemic): Beta-Blockers (Beta1 Selective) may decrease therapeutic effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates): Beta-Blockers may increase vasoconstricting effects of Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates). *Risk C: Monitor*

Etilefrine: Beta-Blockers may decrease therapeutic effects of Etilefrine. Etilefrine may increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

Etofylline: Beta-Blockers may decrease therapeutic effects of Etofylline. *Risk X: Avoid*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fexinidazole: Bradycardia-Causing Agents may increase arrhythmogenic effects of Fexinidazole. *Risk X: Avoid*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Flunarizine: May increase therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Grass Pollen Allergen Extract (5 Grass Extract): Beta-Blockers may increase adverse/toxic effects of Grass Pollen Allergen Extract (5 Grass Extract). More specifically, Beta-Blockers may inhibit the ability to effectively treat severe allergic reactions to Grass Pollen Allergen Extract (5 Grass Extract) with epinephrine. Some other effects of epinephrine may be unaffected or even enhanced (e.g., vasoconstriction) during treatment with Beta-Blockers. Management: Consider alternatives to either grass pollen allergen extract (5 grass extract) or beta-blockers in patients with indications for both agents. Canadian product labeling specifically lists this combination as contraindicated. *Risk D: Consider Therapy Modification*

Herbal Products with Blood Pressure Increasing Effects: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Indoramin: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Isocarboxazid: May increase antihypertensive effects of Antihypertensive Agents. *Risk X: Avoid*

Isoproterenol: Beta-Blockers may decrease therapeutic effects of Isoproterenol. *Risk C: Monitor*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Lacosamide: Bradycardia-Causing Agents may increase AV-blocking effects of Lacosamide. *Risk C: Monitor*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Loop Diuretics: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Mavacamten: Beta-Blockers may increase adverse/toxic effects of Mavacamten. Specifically, negative inotropic effects may be increased. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Methacholine: Beta-Blockers may increase adverse/toxic effects of Methacholine. *Risk C: Monitor*

Methylphenidate: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Mivacurium: Beta-Blockers may increase therapeutic effects of Mivacurium. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

NIFEdipine (Systemic): May increase hypotensive effects of Beta-Blockers. NIFEdipine (Systemic) may increase negative inotropic effects of Beta-Blockers. *Risk C: Monitor*

NIFEdipine (Topical): May increase adverse/toxic effects of Beta-Blockers. *Risk C: Monitor*

Nitrendipine: May increase therapeutic effects of Beta-Blockers. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Topical): May decrease therapeutic effects of Beta-Blockers. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents: May decrease antihypertensive effects of Beta-Blockers. *Risk C: Monitor*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Opipramol: Beta-Blockers may increase serum concentrations of Opipramol. Opipramol may increase serum concentrations of Beta-Blockers. *Risk C: Monitor*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Perazine: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Periciazine: Antihypertensive Agents may increase orthostatic hypotensive effects of Periciazine. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Prazosin: Antihypertensive Agents may increase hypotensive effects of Prazosin. *Risk C: Monitor*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Reserpine: May increase hypotensive effects of Beta-Blockers. Reserpine may increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

Rivastigmine: May increase bradycardic effects of Beta-Blockers. Management: Consider alternatives to the use of rivastigmine and beta-blockers when possible, as product labeling states that the concomitant use of these agents is not recommended. If combined, monitor for bradycardia and syncope. *Risk D: Consider Therapy Modification*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Succinylcholine: Beta-Blockers may increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Tasimelteon: Beta-Blockers may decrease therapeutic effects of Tasimelteon. Management: Consider avoiding nighttime administration of beta-blockers during tasimelteon therapy due to the potential for reduced tasimelteon efficacy. *Risk D: Consider Therapy Modification*

Terazosin: Antihypertensive Agents may increase hypotensive effects of Terazosin. *Risk C: Monitor*

Theodrenaline: May increase bradycardic effects of Beta-Blockers. Beta-Blockers may decrease therapeutic effects of Theodrenaline. *Risk C: Monitor*

Theophylline Derivatives: Beta-Blockers (Beta1 Selective) may decrease bronchodilatory effects of Theophylline Derivatives. *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Urapidil: Antihypertensive Agents may increase hypotensive effects of Urapidil. *Risk C: Monitor*

White Birch Allergen Extract: Beta-Blockers may increase adverse/toxic effects of White Birch Allergen Extract. Specifically, beta-blockers may reduce the effectiveness of beta-agonists that may be required to treat systemic reactions to white birch allergen extract. *Risk X: Avoid*

Zuclopenthixol: May increase or decrease antihypertensive effects of Beta-Blockers. *Risk X: Avoid*

#### Pregnancy Considerations

Exposure to esmolol may cause fetal bradycardia which may continue after esmolol is discontinued. If maternal use of a beta-blocker is needed, fetal growth should be monitored during pregnancy and the newborn should be monitored for 48 hours after delivery for bradycardia, hypoglycemia, and respiratory depression (ESC [Regitz-Zagrosek 2018]).

Esmolol is a short-acting beta-blocker and not indicated for the chronic treatment of hypertension; however, use may be considered as an alternative agent for hypertensive emergencies in pregnancy (ACOG 2019; ESC [Cífková 2020]). Agents other than esmolol may be preferred for the treatment of supraventricular tachycardia, atrial fibrillation, atrial flutter, and ventricular tachycardia in pregnancy. Consult current guidelines for indication specific recommendations (ACC/AHA/HRS [Page 2016]; ESC [Regitz-Zagrosek 2018]).

#### Breastfeeding Considerations

It is not known if esmolol is present in breast milk.

Due to the potential for serious adverse reactions in the breastfed infant, the manufacturer recommends a decision be made whether to discontinue breastfeeding or the drug, considering the importance of treatment to the mother. The short half-life and the fact that it is not intended for chronic use should limit any potential exposure to the breastfeeding infant.

#### Monitoring Parameters

BP, mean arterial pressure, heart rate, continuous ECG, respiratory rate, IV site; serum potassium (especially with kidney impairment); consult individual institutional policies and procedures.

#### Mechanism of Action

Class II antiarrhythmic: Competitively blocks response to  $\beta_1$ -adrenergic stimulation with little or no effect of  $\beta_2$ -receptors except at high doses, no intrinsic sympathomimetic activity, no membrane stabilizing activity

#### Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Beta-blockade: IV: 2 to 10 minutes (quickest when loading doses are administered).

Duration of hemodynamic effects: 10 to 30 minutes; prolonged following higher cumulative doses, extended duration of use.

Distribution:  $V_d$ :

Children  $\geq 2.5$  years and Adolescents  $\leq 16$  years:  $2 \pm 1.4$  L/kg (range: 0.5 to 3.6 L/kg) (Wiest 1991).

Adults: Esmolol:  $\sim 3.4$  L/kg; Acid metabolite:  $\sim 0.4$  L/kg.

Protein binding: Esmolol: 55%; Acid metabolite: 10%.

Metabolism: In blood by red blood cell esterases; forms acid metabolite (negligible activity; produces no clinically important effects) and methanol (does not achieve concentrations associated with methanol toxicity).

Half-life elimination:

Children  $\geq 18$  months and Adolescents  $\leq 16$  years: Variable; mean range: 2.7 to 4.8 minutes (reported full range: 0.2 to 9.9 minutes) (Adamson 2006; Cuneo 1994; Tabbutt 2008a; Wiest 1991; Wiest 1998).

Adults: Esmolol: 9 minutes; Acid metabolite: 3.7 hours; elimination of metabolite decreases with end-stage kidney disease.

Excretion: Urine ( $\sim 73\%$  to  $88\%$  as acid metabolite,  $<2\%$  unchanged drug).

#### Patient Counseling Points

(For additional information see "Esmolol: Patient drug information")

#### **What is this drug used for?**

- It is used to treat a fast heartbeat.
- It is used to treat high blood pressure.
- It may be given to you for other reasons. Talk with the doctor.

**All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:**

- Nausea

**WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:**

- High potassium like abnormal heartbeat, confusion, dizziness, passing out, weakness, shortness of breath, or numbness or tingling feeling.
- Acidosis like confusion, fast breathing, fast heartbeat, abnormal heartbeat, severe abdominal pain, nausea, vomiting, fatigue, shortness of breath, or loss of strength and energy.
- Dizziness
- Passing out

- Sweating a lot
- Shortness of breath
- Excessive weight gain
- Swelling of arms or legs
- Slow heartbeat
- Abnormal heartbeat
- Injection site burning, redness, pain, swelling, or severe irritation
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

**Note:** This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

**Consumer Information Use and Disclaimer:** This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Esobloc;
- (AR) Argentina: Brevibloc | Esmolol richet;
- (AT) Austria: Brevibloc | Esmolol Amomed | Esmolol hikma;
- (AU) Australia: Brevibloc | Esmolol HCL | Esmolol juno;
- (BR) Brazil: Brevibloc | Brevibloc diluido;
- (CH) Switzerland: Brevibloc | Esmolol Orpha;
- (CN) China: Esmolol | Xin luo ping;
- (CO) Colombia: Brevibloc | Esmoblock;
- (CZ) Czech Republic: Brevibloc | Esmocard;
- (DE) Germany: Brevibloc | Esmocard | Esmolol | Esmololhydrochlorid Noridem;
- (EE) Estonia: Esmocard;
- (EG) Egypt: Brevibloc premixed;
- (ES) Spain: Brevibloc;
- (ET) Ethiopia: Esmobloc;
- (FI) Finland: Brevibloc;
- (FR) France: Brevibloc | Esmocard;
- (GB) United Kingdom: Brevibloc | Esmolol;
- (GR) Greece: Brevibloc | Esmocard;
- (HK) Hong Kong: Brevibloc;
- (HU) Hungary: Brevibloc | Esmocard;
- (IE) Ireland: Brevibloc;
- (IN) India: Esocard;

- (IT) Italy: Brevibloc | Esmocard | Esmololo cloridrato Hikma;
- (JP) Japan: Brevibloc;
- (KR) Korea, Republic of: Jeil brevibloc;
- (KW) Kuwait: Brevibloc | Esmolol HCL;
- (LT) Lithuania: Brevibloc | Esmocard | Esmolol;
- (LV) Latvia: Brevibloc | Brevibloc | Esmocard;
- (MX) Mexico: Crevimol | Esmolol tecnofarma | Nalopres | Tamgho;
- (MY) Malaysia: Brevibloc | Esmolol | Esocard;
- (NL) Netherlands: Brevibloc | Esmolol HCL;
- (NO) Norway: Brevibloc | Esmolol baxter;
- (NZ) New Zealand: Brevibloc | Esmolol;
- (PH) Philippines: Brevibloc;
- (PL) Poland: Brevibloc | Esmocard;
- (PR) Puerto Rico: Brevibloc | Esmolol | Esmolol HCL;
- (PT) Portugal: Brevibloc | Esmocard | Esmolol hikma;
- (QA) Qatar: Brevibloc | Brevibloc Premixed | Esmobloc;
- (RO) Romania: Brevibloc;
- (RU) Russian Federation: Brevibloc | Brevicard;
- (SA) Saudi Arabia: Brevibloc | Brevibloc premixed;
- (SE) Sweden: Brevibloc;
- (SG) Singapore: Brevibloc;
- (SI) Slovenia: Brevibloc | Esmocard;
- (SK) Slovakia: Esmocard;
- (TH) Thailand: Esbloc | Esmolol;
- (TR) Turkey: Brevibloc | Esmobloc | Kardoes Premix | Multiflex ermoloc premiks;
- (TW) Taiwan: Brevibloc | Esmolol HCL;
- (UA) Ukraine: Bibloc;
- (UY) Uruguay: Dublon | Dublon fuerte;
- (ZA) South Africa: Brevibloc | Esmocard

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